

Debriefings in Misinformation Research Should Include Fact-Checks to Reduce False Beliefs and Increase Perceived Learning

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Abstract

Debriefings are a cornerstone of ethical research, especially when deception is used. In (experimental) misinformation research, this often involves exposing participants to false claims without their awareness. Debriefings both disclose deception and act as interventions, warning participants about misinformation. Yet, unlike such interventions (e.g., fact-checking, media literacy tools) they reveal that researchers deliberately introduced the falsehoods, raising concerns about unintended effects on trust and skepticism. In a pre-registered U.S.-based experiment (N = 896), we compare the effects of no debriefing, a standard debriefing, and a specific debriefing that includes fact-checks and source links. We find that only the specific debriefing significantly reduced belief in false news. The standard debriefing slightly reduced belief in true news, suggesting that explicit warnings may heighten generalized skepticism. Both debriefings improved participants' attitudes toward the study, increasing perceived transparency and reduced feelings of manipulation. These findings suggest that while general warnings may not reduce misperceptions and may come with small unintended side effects, specific debriefings can correct false beliefs and foster trust — supporting their use as an ethical and effective practice in misinformation research.

Keywords: Debriefings, Ethics, Misinformation, Trust, Skepticism, Experiments, Deception

1. Introduction

Experimental research frequently employs deception, such as withholding information from participants or exposing them to false claims without revealing that the claims are false. This ensures the internal validity of the experiment and the successful delivery of the treatment (Holmes, 1976; Walster et al., 1967). One prevalent form of deception in misinformation research, central to this study, involves presenting misinformation to participants without their awareness (Clayton et al., 2024; Greene et al., 2023). Given ethical obligations to correct participants' induced false beliefs, effective debriefing becomes essential, not only to correct misperceptions but also to preserve participants' trust and willingness to engage in future research (Oczak and Niedźwieńska, 2007).

However, recent research suggests that interventions correcting misinformation, though effective in reducing false beliefs, can inadvertently increase skepticism toward accurate information and erode trust in credible sources (Hoes et al., 2024; Altay et al., 2024; Thorson, 2024; van der Meer et al., 2023; van Erkel et al., 2024). For instance, Hoes et al. (2024) show that common misinformation interventions (e.g., fact-checking, media-literacy messaging) often prime individuals to approach all information skeptically, diminishing trust in accurate information and that warnings about misinformation can decrease trust in scientists (Hoes et al., 2025). Debriefings in experimental research may produce similar effects: not only do they

explicitly warn participants about the presence of misinformation, but they also reveal that participants were deliberately deceived. This combination could amplify feelings of being misled, increasing skepticism toward the study or science in general. At the same time, as Altay et al. (2024) and van Erkel et al. (2024) note, transparency-focused interventions can promote trust when they clearly signal reliability and distinguish false from true content. Debriefings, particularly when specific and detailed, may therefore also enhance perceptions of researcher transparency and foster greater trust and engagement.

Given these nuanced findings, this pre-registered study systematically investigates both potential positive and negative consequences of specific and general debriefing strategies following exposure to misinformation within the study. In line with recent research on debriefings Clayton et al. (2024), we operationalize the difference between a specific and general debriefing such that in the former the false news items used in the study are actively debunked and accompanied by links to fact-checking websites, whereas in the latter they are not. Figure 1 depicts the two debriefings.

Based on previous evidence (Clayton et al., 2024), we hypothesized that specific debriefings will correct false beliefs more effectively than general debriefings. Specifically, we examine belief accuracy for false news that were presented both before and after treatment, as well as for novel false news items shown only post-treatment. The key difference here is that the repeated news items were corrected in the specific debriefing, whereas the novel news items were not. In addition, we explore a series of open-ended research questions regard-

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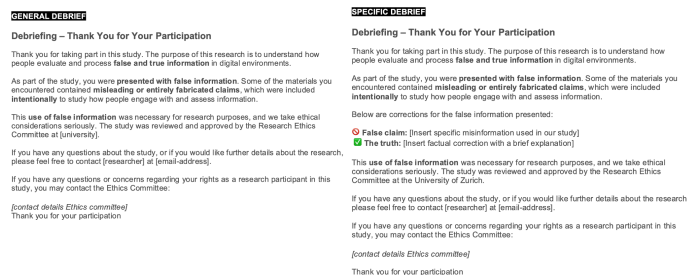


Figure 1: Debriefings used as the Treatment Materials. The General Debriefing is depicted on the left. The Specific Debriefing on the right.

ing broader outcomes that remain largely unexamined in prior work on debriefings. While earlier studies have primarily focused on belief correction as the main outcome (Clayton et al., 2024; Greene et al., 2023; Ross et al., 1975; Walster et al., 1967), few have systematically investigated how debriefings affect trust in information, perceptions of transparency, or willingness to participate in future research. Though some have raised theoretical concerns about such consequences Oczak and Niedźwieńska (2007), empirical evidence remains limited. We therefore assess whether debriefings unintentionally heighten skepticism toward accurate information (using both repeated and novel news items), or conversely, enhance trust and attitudes toward the study. By simultaneously examining potential harms (e.g., generalized skepticism) and benefits (e.g., trust and transparency), our study offers novel insights into how debriefings can better balance methodological rigor, ethical responsibility, and participant welfare in experimental misinformation research.

2. Methods

The pre-registrations, materials, R scripts, and the data to reproduce the findings are publicly available on OSF.

On June 13, 2025, we recruited 900 U.S. participants via the platform Prolific. After excluding four participants who failed the attention check, our final sample consisted of 896 respondents (median age = 37; 482 women and 404 men; median education = college degree). Before being randomly assigned to one of the two treatment groups (specific or general debriefing) or the control group (no debriefing), participants completed a set of demographic questions and a pre-treatment survey in which they rated several true and false news items and indicated their attitudes toward the study, the researchers, and science more broadly. After the treatment, participants again answered questions — some of which repeated earlier items, and others that were new, as detailed below. For ethical reasons, all participants received a final debriefing at the end of the study, regardless of condition.

We selected a total of six false news items and six true news items. Of these six, four were part of the pre- and post-treatment survey (repeated), and the two additional items were presented post-treatment only (‘novel’ items). The false news items were selected from a larger pool of headlines that were gathered by

visiting the prominent fact-checking websites Snopes and PolitiFact. The true news items were selected by visiting prominent legacy media outlets such as the New York Times, the Guardian and the Washington Post. Of all the headlines, we selected six true and six false news items balanced in terms of political leaning (i.e., pro Republican or pro Democrat). We based this approach on current best practices in the field Pennycook et al. (2021). The survey, including the true and false news items, can be found in on OSF.

For the true and false news items, and all the questions below, participants answered on a five-point Likert scale (‘Strongly disagree’, ‘Disagree’, ‘Neutral’, ‘Agree’, ‘Strongly agree’).

Attitudes towards the study were measured only post-treatment. We did not measure them pre-treatment because participants could not meaningfully evaluate the study’s purpose, transparency, or ethical aspects before receiving the debriefing. Measuring them beforehand could also have primed participants or drawn unnecessary attention to the design of the study. Participants indicated whether they agreed with eight statements about the study. Five statements were positive (e.g., “The way this study was conducted was ethically acceptable”) and four were negative (“I felt manipulated by how this study was designed”). After reverse coding the negative items, the scale has high internal validity (Cronbach’s alpha = 0.78).

Trust in scientists was measured before and after the treatment. Participants indicated whether they agreed with six statements about scientists, three were positive (e.g., “I trust scientists to act responsibly and in good faith”) and three were negative (“I am generally skeptical about the intentions of scientists”). After reverse coding the negative items, the scale has high internal validity (Cronbach’s alpha = 0.85 pre-treatment and 0.86 post-treatment).

Trust in institutions was measured pre- and post-treatment. Participants indicated whether they agreed with four statements about news organizations, fact-checkers, and the government, such as “I trust news organizations to provide accurate and balanced information”. The scale has low internal validity (Cronbach’s alpha of 0.59 pre-treatment and 0.66 post-treatment). On OSF, we break down the scale and show that the treatments did not affect any of the specific items of the scale.

As with attitudes toward the study, willingness to participate in future studies was measured only post-treatment, since participants could not meaningfully assess their willingness to participate in similar studies before knowing the study’s purpose and procedure. Participants indicated whether they agreed with the following statements “I am willing to participate in future research studies” and “I would consider taking part in a similar research study that includes exposure to false or misleading information” (Cronbach’s alpha = 0.67). On OSF, we break down the scale and show that the treatments did not affect any of the specific items of the scale.

Statistical Analyses

In all the models, we control for age, gender, education, and political orientation. We report the unstandardized beta (b) in the text, while in the Figures we report the standardized beta to

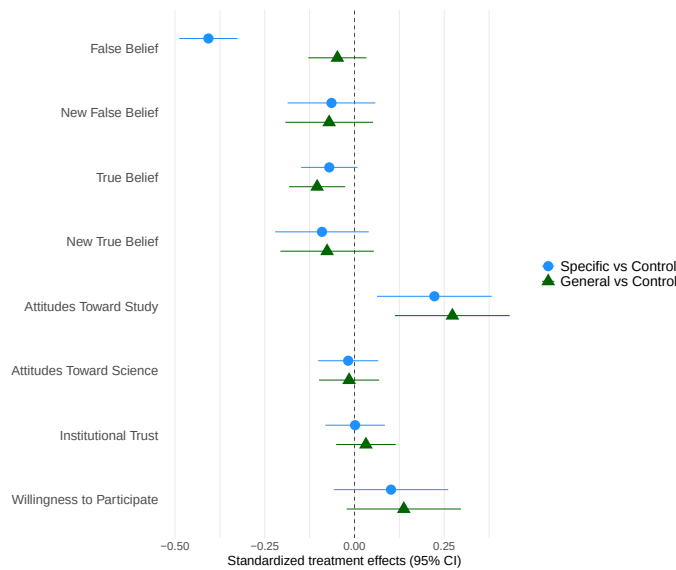


Figure 2: Main effects of the debriefings on all the dependent variables.

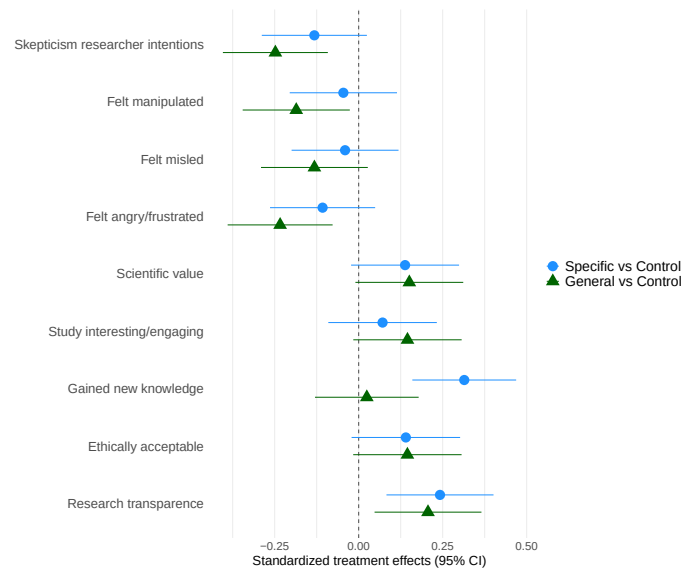


Figure 3: Main effects of the debriefings on attitudes toward the study.

facilitate comparison across variables with different scales. We report all the regression tables on OSF.

To estimate the effect of the debriefings on the perceived accuracy of true and false news, we analyse the data at the response level and conduct linear mixed-effect models with random effects for news item and participant. For news items measure pre- and post-treatment, we interact condition with a term for pre/post, whereas for the novel news items, only measured post-treatment, we do not.

For all the other outcomes, we use ordinary least squares (OLS) linear regressions. For outcomes measured pre- and post-treatment, we estimate differences post-treatment while controlling for pre-treatment levels, whereas for outcomes measured only post-treatment, we do not.

3. Results

In Figure 2, we plot the effect of the Debriefings. The blue circles compare the Specific Debriefing to the Control, while the green triangles compare the General Debriefing to the Control.

The debriefings will reduce belief in false news

Compared to the Control Condition, the Specific Debriefing reduced belief in false news by 0.78 points on the 7-point scale ($p < .001$), while the General Debriefing did not ($b = 0.09$, $p = .25$). The effect of the Specific Debriefing was 0.69 points larger ($p < .001$) than that of the General Debriefing. This suggests that general warnings about false news do not reduce belief in false news whereas specific fact-checks do.

As can be seen in Figure 2, none of the debriefings significantly reduced belief in novel false news, and the effect sizes are of similar magnitude (0.13 points for the General Debriefing and 0.12 points for the Specific Debriefing).

Effects of the debriefing on true news

Compared to the Control Condition, the Specific Debriefing reduced belief in true news by 0.20 points on the 7-point scale ($p = .009$), while the General Debriefing did not ($b = 0.14$, $p = .08$). However, the effect of the debriefings are not significantly different from each other ($b = 0.07$, $p = .40$). This suggests that this negative effect on true news is mostly driven by shared properties of the debriefings (i.e., the general warning), not the fact-checks. We further reflect on this in the Discussion.

As can be seen in Figure 2, none of the debriefings significantly reduced belief in novel true news, and the effect sizes are of similar magnitude (0.15 points for the General Debriefing and 0.18 points for the Specific Debriefing).

Effects of the debriefing on attitudes toward the study

Compared to the Control Condition, the Specific Debriefing ($b = 0.13$, $p = .006$) and the General Debriefing ($b = 0.16$, $p = .001$) improved participants' (positive) attitudes toward the study. In Figure 3, we break down the results by specific attitudes towards the study. Compared to the Control and the General Debriefing, the Specific Debriefing increased participants' feeling that they gained new knowledge by participating in the study ($b = 0.31$, $p < .001$). Compared to the control, the General Debriefing ($b = 0.21$, $p = .011$) and the Specific Debriefing ($b = 0.24$, $p = .003$) increased the perceived transparency of the study. Compared to the control, the general debriefing reduced skepticism of researchers' intentions ($b = 0.25$, $p = .002$), feeling of being manipulated ($b = 0.19$, $p = .022$), and anger or frustration associated with the study ($b = 0.23$, $p = .003$) — these effects are not statistically different from the Specific Debriefing.

Effects of the debriefing on attitudes toward scientists

None of the debriefings significantly influenced attitudes towards scientists (e.g., "I trust scientists to act responsibly and

in good faith”), both effect sizes are extremely small ($b = 0.01$, $ps > .69$).

Effects of the debriefing on institutional trust

None of the debriefings significantly influenced institutional trust (i.e., trust in news organizations, fact-checkers, and the government), both effect sizes are very small ($b < 0.04$, $ps > .47$).

Effects of the debriefing on willingness to participate in surveys

None of the debriefings significantly influenced participants’ willingness to participate in future surveys ($b < 0.09$, $ps > .08$). On OSF, we break down the scale and show that the treatments did not affect any of the specific items of the scale.

Moderating effect of total number of survey taken on Prolific

We found that the effect of the specific debriefing on false belief was stronger among participants who have taken more surveys on Prolific ($b = 0.09$, $p = .032$). Moreover, the positive effects of the Specific Debriefing ($b = 0.05$, $p = .029$) and the General Debriefing ($b = 0.06$, $p = .013$) on attitudes towards the study were stronger among participants who have taken more surveys on Prolific. We found no other statistically significant differences. We expected the effect of the debriefing to be weaker, not stronger, among participants who have more experience with surveys.

4. Conclusion & Discussion

This study offers a comprehensive assessment of how different debriefing strategies affect participants in experimental misinformation research. While prior work has mainly focused on belief correction, we extend this literature by examining not only whether debriefings effectively correct misperceptions, but also how they shape participants’ attitudes toward the study, trust in science, willingness to participate in future research, as well as their belief in true information. By investigating both intended and unintended consequences—including the possibility that debriefings heighten generalized skepticism—we provide a more complete picture of how debriefings function in the context of misinformation studies. Focusing on the increasingly common practice of exposing participants to misinformation without their awareness, we compare the effects of a general warning and a specific corrective debriefing. Our findings offer both reassuring and cautionary insights for researchers navigating the ethical and methodological trade-offs involved in the use of deception.

Consistent with prior work on fact-checking and debunking interventions (Clayton et al., 2024), specific debriefings improved belief accuracy regarding the exact falsehoods presented during the study. In contrast, general debriefings — though ethically compliant — did not reduce belief in false information. This finding reinforces the growing consensus that vague or broad warnings about misinformation are insufficient to correct misperceptions and that specific, claim-level corrections remain a best practice when addressing false beliefs.

Yet our findings also reveal potential unintended consequences. The standard debriefing slightly reduced belief in true news that participants had seen before, suggesting that even ethically motivated transparency can prime generalized skepticism. While this is in line with recent concerns that interventions aimed at debunking misinformation can erode trust in accurate content or sources (Hoes et al., 2024; Thorson, 2024; Altay et al., 2024), our findings refine that concern. Specifically, the similarity in effect size across the general and specific debriefing conditions suggests that this reduction in belief in true news is not primarily driven by the act of debunking itself, but by shared elements of both debriefings — namely, the warning about misinformation and the revelation of intentional deception. Our design cannot isolate which component drives this effect, but the pattern suggests that warning participants about misinformation — especially when framed as having been deliberately introduced by the researchers — may undermine confidence in adjacent or truthful content.

Importantly, however, our findings also point to positive consequences. Both debriefings increased perceived transparency and improved attitudes toward the study. The standard debriefing also lowered feelings of manipulation and frustration. These results echo recent work suggesting that interventions can foster trust when they are perceived as transparent and clearly delineate truth from falsehood (Altay et al., 2024; van Erkel et al., 2024). Compared to the standard debriefing, the specific debriefing was particularly effective in helping participants feel they had gained new knowledge, further supporting the idea that transparent, claim-level corrections can enhance the perceived value of research participation. Contrary to concerns about backlash from deception, we found no negative effects on trust in scientists, institutions, or willingness to participate in future studies.

Taken together, these findings offer a more nuanced understanding of how debriefings function — not merely as ethical afterthoughts but as active interventions with meaningful consequences. While the corrective power of specific debriefings is clear, their broader informational and psychological effects deserve more attention. Future work should explore how different framing choices — e.g., apologetic versus assertive tones, attribution of blame, or source emphasis — may shape these effects. Moreover, experimental designs that allow finer isolation of the warning versus correction components are needed to disentangle their respective roles. Additionally, future studies should include behavioral measures — such as actual participation in a second follow-up survey — to assess attrition and willingness to re-engage more robustly than through self-reports alone.

Ultimately, our results suggest that debriefings can both promote accuracy and foster trust, while also introducing risks of overgeneralized skepticism. Ethical experimental design must therefore strike a careful balance between maximizing transparency and learning, while minimizing confusion and doubt.

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